Jesús M. Rueda-Becerril | Ph.D.

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Professional Summary

Physicist with a Ph.D. in Physics and over ten years of research experience in computational astrophysics, numerical methods, and high-performance computing. One term of classroom instruction at the graduate level and multi-year experience mentoring graduate researchers through the transition from confusion to independent work. Teaching approach built on intuition and critical thinking over memorization, and on collaborative problem-solving.

Teaching & Mentoring Experience

Graduate Substitute Instructor, *Thermodynamics*, Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo. **Jun 2018**

Mentoring, *Graduate student*, Zachary Davis, Department of Physics and Astronomy, Purdue University. **2018 – 2022**

Mentoring, *Graduate student*, Amanda O. Harrison, Department of Physics and Astronomy, Purdue University. **2018 – 2020**

Mentoring, *Graduate student*, Hao Zhang, Department of Physics and Astronomy, Purdue University. **2018 – 2019**

Education

Universitat de València, Spain, *Ph.D. in Physics*, Excellent Cum Laude. **Jul 2017**

Universidad Michoacana de San Nicolás de Hidalgo, Mexico, *M.Sc. in Physics*. **Sep 2011**

Universidad Autónoma del Estado de México, Mexico, *B.Sc. in Physics*. **Dec 2008**

Experience

Mercor, *Independent Contractor*. **Jan 2026 — Present**

- Evaluate physics reasoning in AI model outputs for technical accuracy, identifying failure modes and providing structured feedback.

Purdue University, *Postdoctoral Research Fellow*. **Oct 2018 – Nov 2020**

- Obtained and led a NASA-funded grant studying radiation from active galaxies.
- Mentored three graduate researchers; two are now postdoctoral researchers.
- Published 3 peer-reviewed papers.

Rochester Institute of Technology, *Postdoctoral Research Associate*. **Feb 2021 — Apr 2022**

- Led an NSF-sponsored project upgrading a high-performance simulation code for modeling black hole binaries.
- Collaborated on a multi-institutional study of binary neutron star mergers, resulting in 2 publications.

Universidad Michoacana de San Nicolás de Hidalgo, *Postdoctoral Research Fellow*. **Jan – Sep 2018**

- Developed a Python pipeline to process black hole simulation outputs from the GRTrans numerical code, generating SVM training data for radio image prediction.
- Built open-source data analysis and visualization tools in Python for radiative transfer phenomena in relativistic astrophysics scenarios.

Skills

Physics: Mechanics, Electromagnetism, Thermodynamics, Relativity, Statistical Physics, Astrophysics

Computing: Python, Fortran, C/C++, high-performance/parallel computing

Languages: English (fluent), Spanish (native)

Selected Publications (9 total, 3 first-author)

- Full list available on request.